Clebsch series figure set

This document previews the reusable figures of `papers/clebsch-series-figures`. It is a proof sheet, not a manuscript: each figure is defined in its own `\input`-able file and is meant to be placed by a host paper with its own caption. See `README.md` for the input recipe and for the three claim statuses the drawings encode.

D1 — the series map

The five-paper map, with Paper III on top descending the golden sign into the central torsor, Papers I and II joined to it by horizontal recognition passages, and Papers IV and V below. Two conventions carry meaning. The Paper IV edge is *dashed*: no theorem carries its reconstructed plane to the oriented cubic torsor, so the bridge is an open obligation rather than an established arrow. The Paper V node states an *intent*: that paper is unwritten and the Rosetta theorem is not proved.

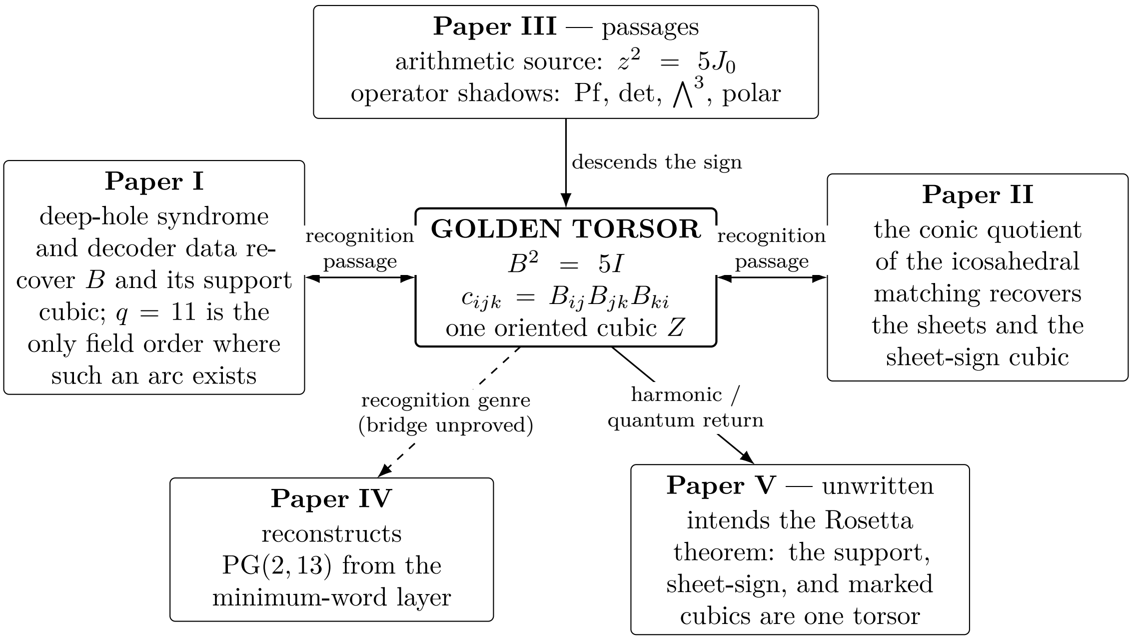


Figure 1: The series map with no node emphasised.

D1 — the five "you are here" variants

Each numbered paper's forward version carries the same map with its own node emphasised by a heavier rule and a light tint.

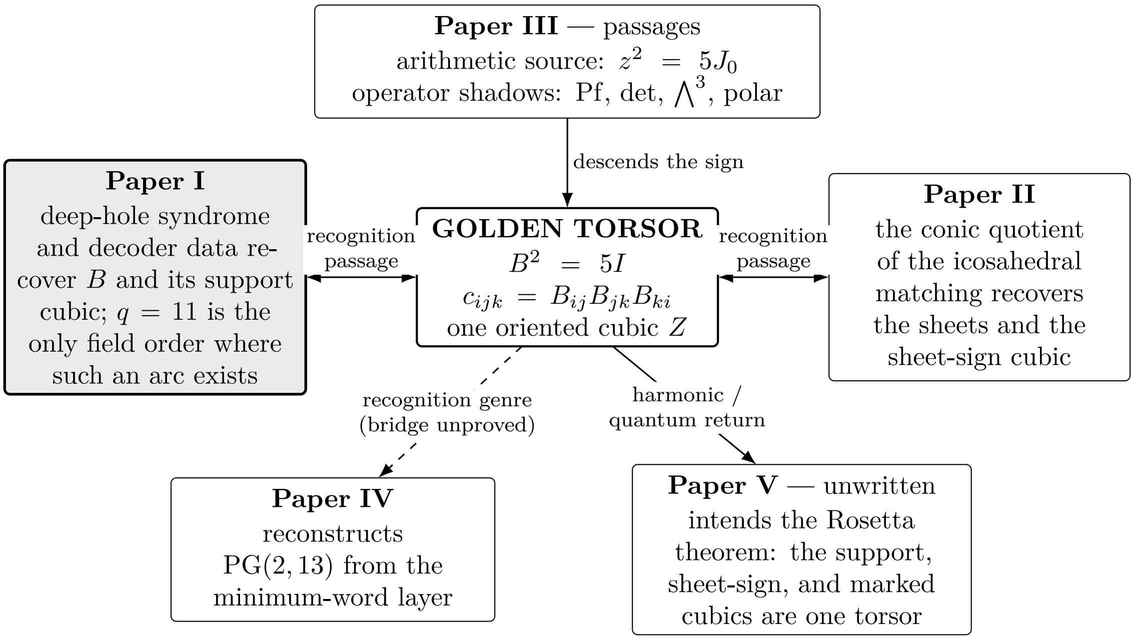


Figure 2: `\ClebschSeriesMap{I}` — for Paper I.

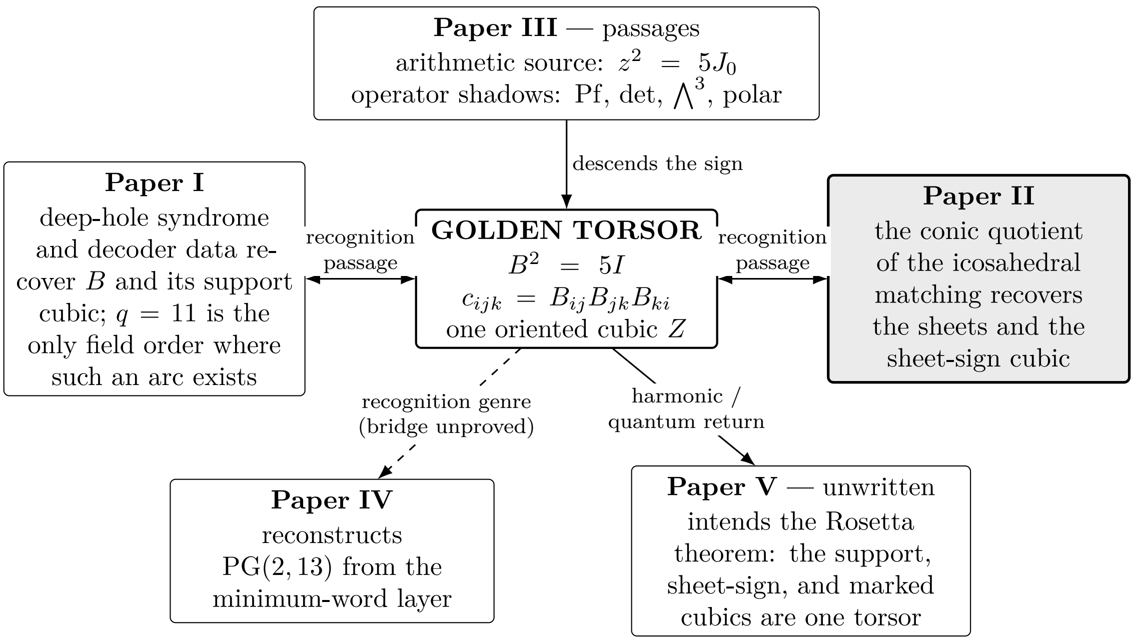


Figure 3: `\ClebschSeriesMap{II}` — for Paper II.

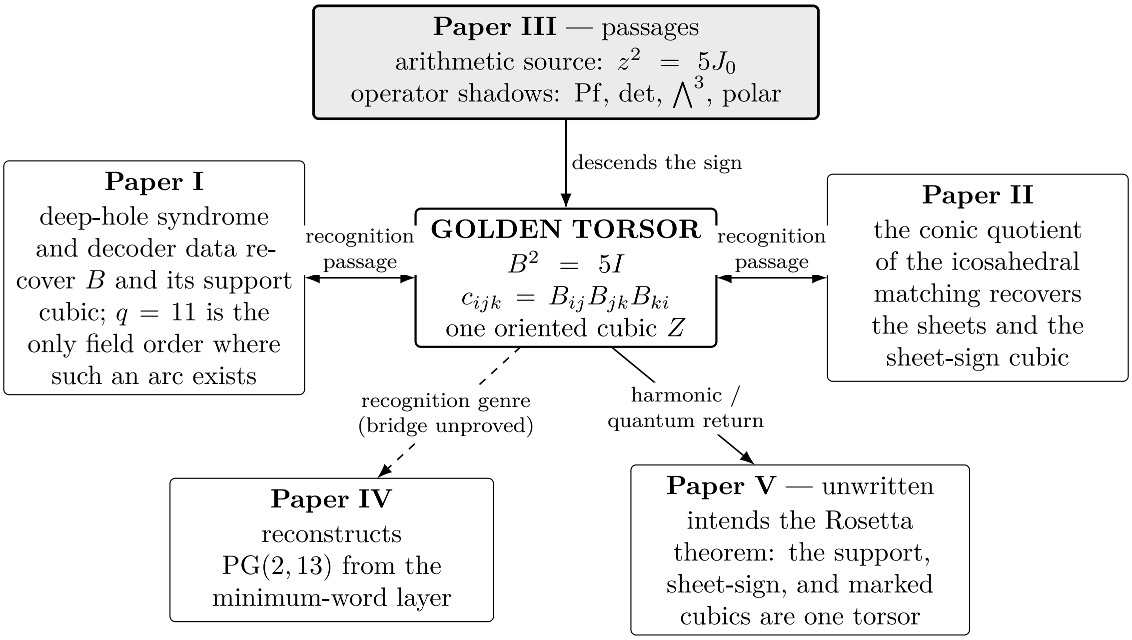


Figure 4: `\ClebschSeriesMap{III}` — for Paper III.

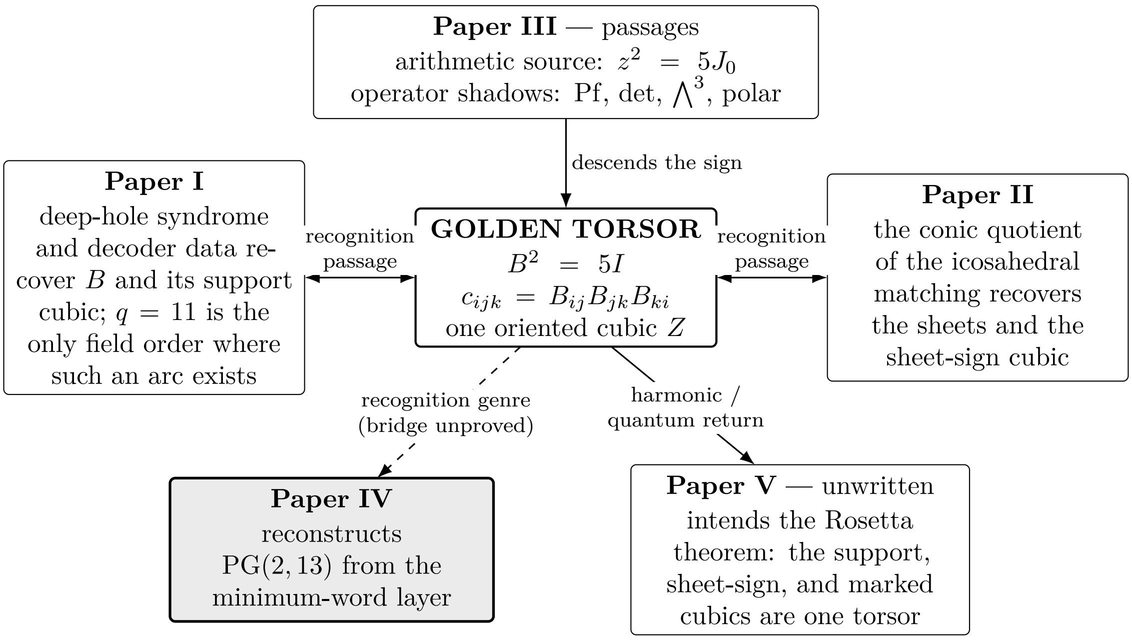


Figure 5: `\ClebschSeriesMap{IV}` — for Paper IV.

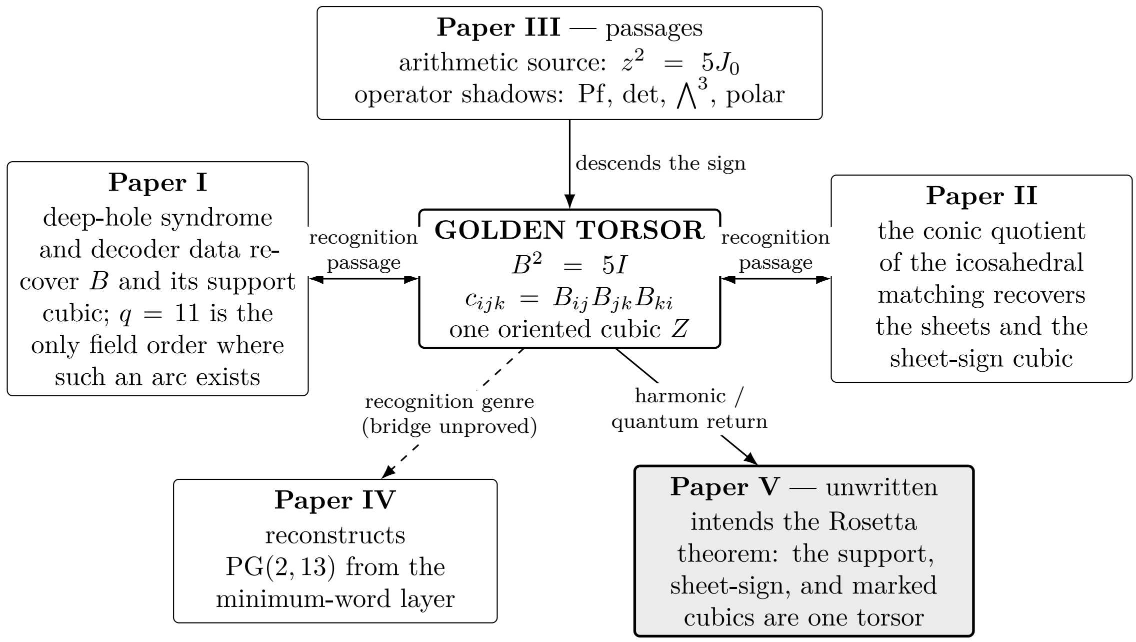
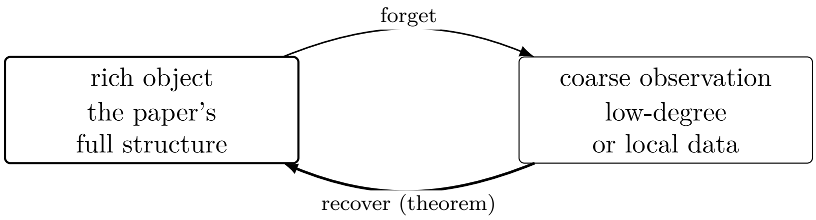


Figure 6: `\ClebschSeriesMap{V}` — for Paper V.

D2 — the recognition template

The programme is a template with instances, not a study of one object. The shape is drawn once; the table instantiates its two slots.



	rich object	*coarse observation*
Paper I	the Clebsch code, a six-arc in $\mathrm{PG}(2, 11)$	its projective deep-hole syndrome locus
Paper II	the icosahedral conic matching and its two sheets	a two-valued strength-two quadratic trade
Paper III	the order-six conference operator C	four-local principal-minor data (the aligned four-sets)
Paper IV	the marked conic plane $\mathrm{PG}(2, 13)$	the minimum-word layer of its passant code
AME paper	a stabilizer $\mathrm{AME}(2m, q)$ state	its product-unitary symmetries

Figure 7: One shape, five instances. The rich object is forgotten down to a coarse observation, and a theorem of the named paper recovers it.

D3 — the quantifier ladder

Each bar is a theorem's quantifier range. Where the theorem's answer is a single exceptional object, it is marked as a distinguished point at the end of the general range. The visual argument is general hypothesis, exceptional conclusion.

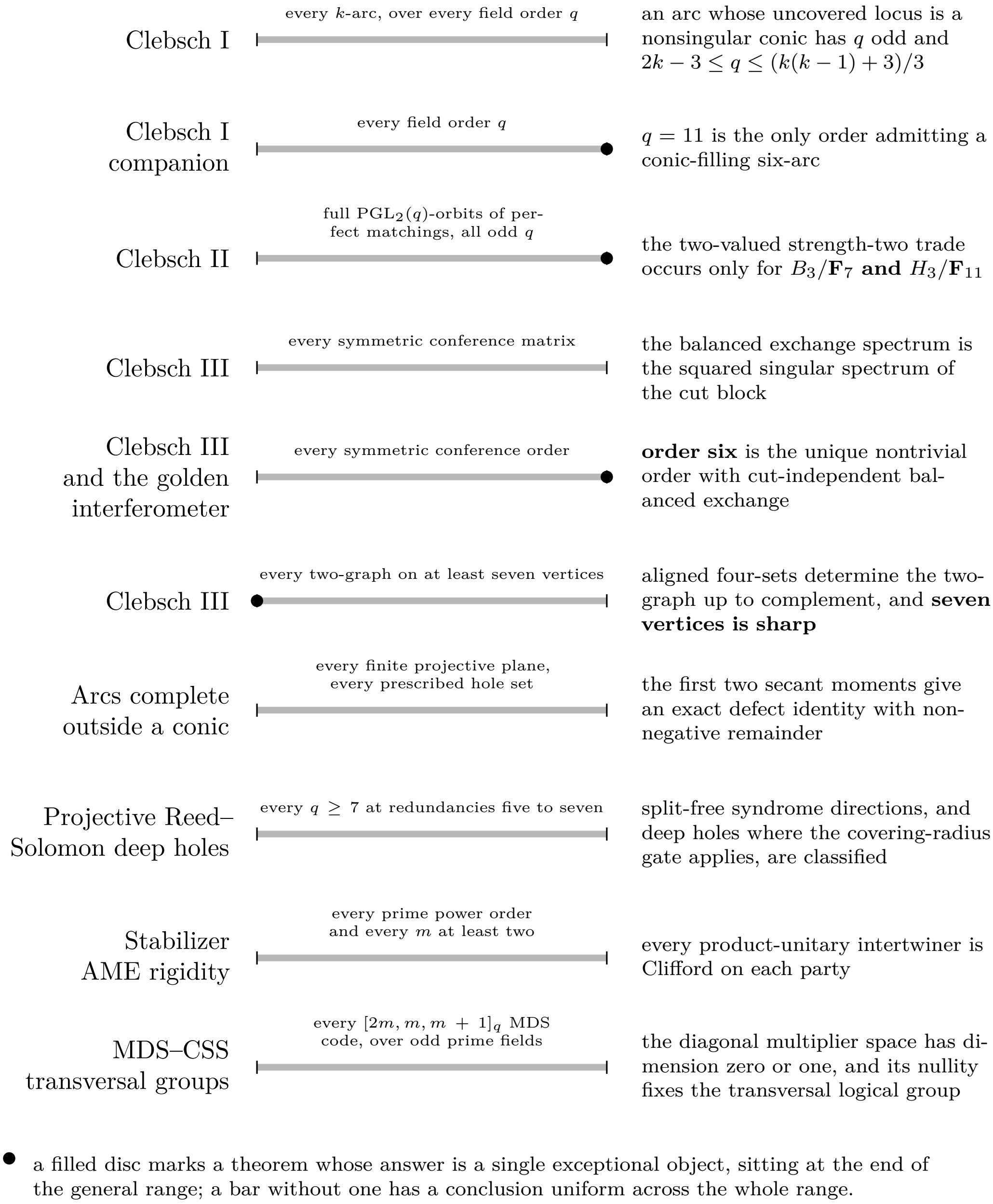


Figure 8: Quantifier ranges of the portfolio's general theorems.

D4 — the four shadows, billed by type

A redesign of Paper III's four-shadow figure. The four descriptions are not four coincidences of equal standing, and the diagram now separates three registers.

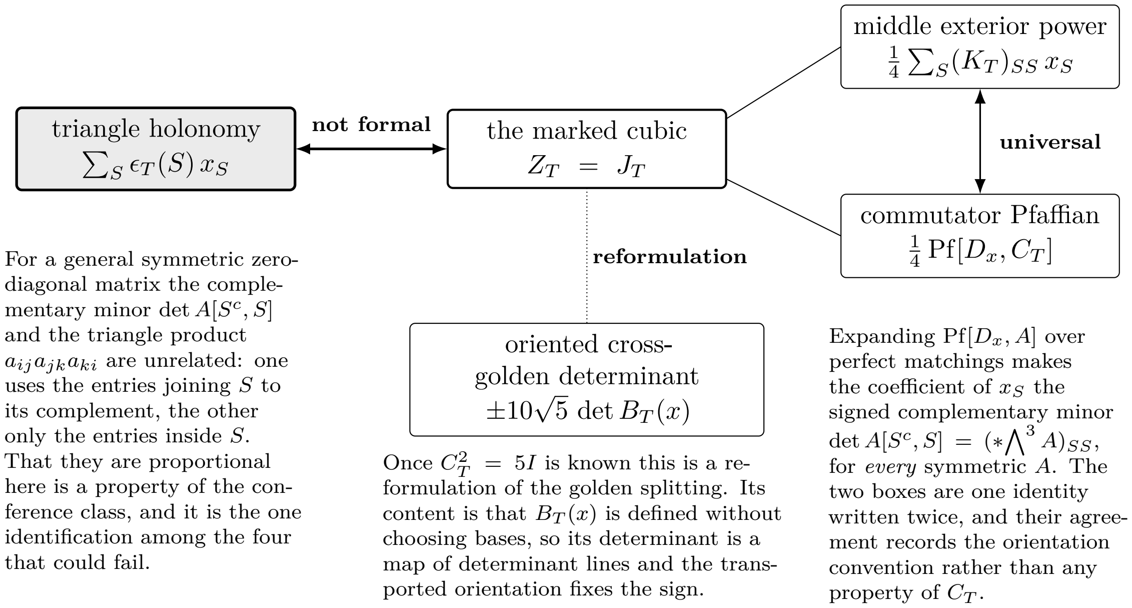


Figure 9: The four descriptions of the marked cubic, separated by what kind of statement each agreement is.

D5 — the determinantal ladder for a Seidel matrix

The reconstruction theorem deliberately consults the weakest rung.

order 1. $\det C[\{i\}] = 0$. Every first-order principal minor vanishes: the diagonal is zero. No information.

order 2. $\det C[\{i,j\}] = -1$, constantly. No information.

order 3. $\det C[\{i,j,k\}] = 2\,\epsilon(ijk)$, twice the triangle sign. The full third-order list hands over every triangle product, so reconstruction up to switching is immediate: this rung trivializes the problem.

strictly stronger data.
The theorem consults none of it.

order 4. $\det C[Q] = 3 - 2w(Q) \in \{-3, 5\}$: one bit per four-set, and invariant under $C \mapsto -C$. This is the data the reconstruction theorem uses. It determines C up to switching and global negation, it needs at least seven vertices — and seven is sharp — and $O(n^2)$ selected determinants suffice.

the weakest rung, and the one the theorem uses

Figure 10: Principal minors of a Seidel matrix by order, and which rung the reconstruction theorem uses.

D6 — Paper III's causal spine

The spine is the same in both variants; only the return arrow differs. The `current` variant returns by the reconstruction theorem the manuscript proves today. The `final` variant returns by the triangle–Pfaffian recognition, which is proved but not yet promoted into the manuscript — a third claim status, drawn as a *double* rule with a tag naming it. It is neither of D1's two caveats: the dashed Paper IV edge there marks a bridge no theorem supports, and the Paper V node marks a paper that does not exist. Nothing here is speculative; the double rule says forthcoming.

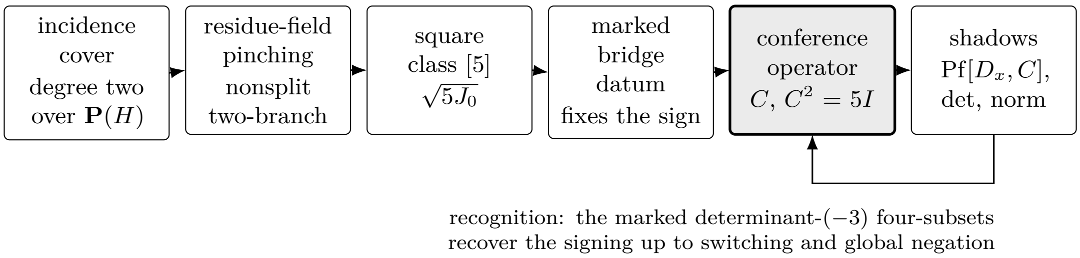
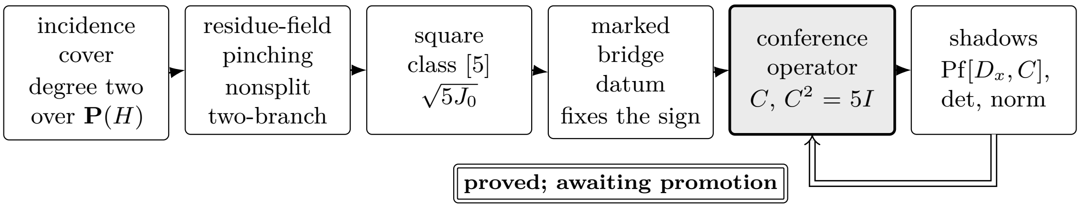


Figure 11: `\ClebschCausalSpine[current]`, the default. From the incidence cover to the operator's shadows, returning by the reconstruction theorem the manuscript states: marked determinant-(-3) four-subsets recover the signing up to switching and global negation.



recognition, stronger form: for symmetric A with zero diagonal and *every* off-diagonal entry nonzero, $\mathrm{Pf}[D_x, A] = \mu T_A(x)$ with $\mu \neq 0$ forces $A^2 = \lambda I$; restricted to the scalar sign locus $A = sB$, $b_{ij} = \pm 1$, it recovers the unique order-six conference class together with its two outer orientations $\mu = \pm 4$

The order is a conclusion, not a hypothesis: $\mathrm{Pf}[D_x, A]$ has degree $n/2$ for an operator of order n, while the triangle cubic T_A has degree three, so unhomogenized proportionality is possible only at $n = 6$.

Proved in characteristic $\neq 2$, and formalized over an integral domain: the quadratic converse and the pentagon gauge are kernel-checked, while the orientation split rests on one compiled finite check. It is not yet a statement of the manuscript.

Figure 12: `\ClebschCausalSpine[final]`. The same spine returning by the triangle–Pfaffian recognition instead. The two hypotheses do different work: nonzero off-diagonal entries give $A^2 = \lambda I$ across the whole nonzero-edge torus, and the scalar sign locus is what pins the golden value and produces the order-six conference class with its two outer orientations.